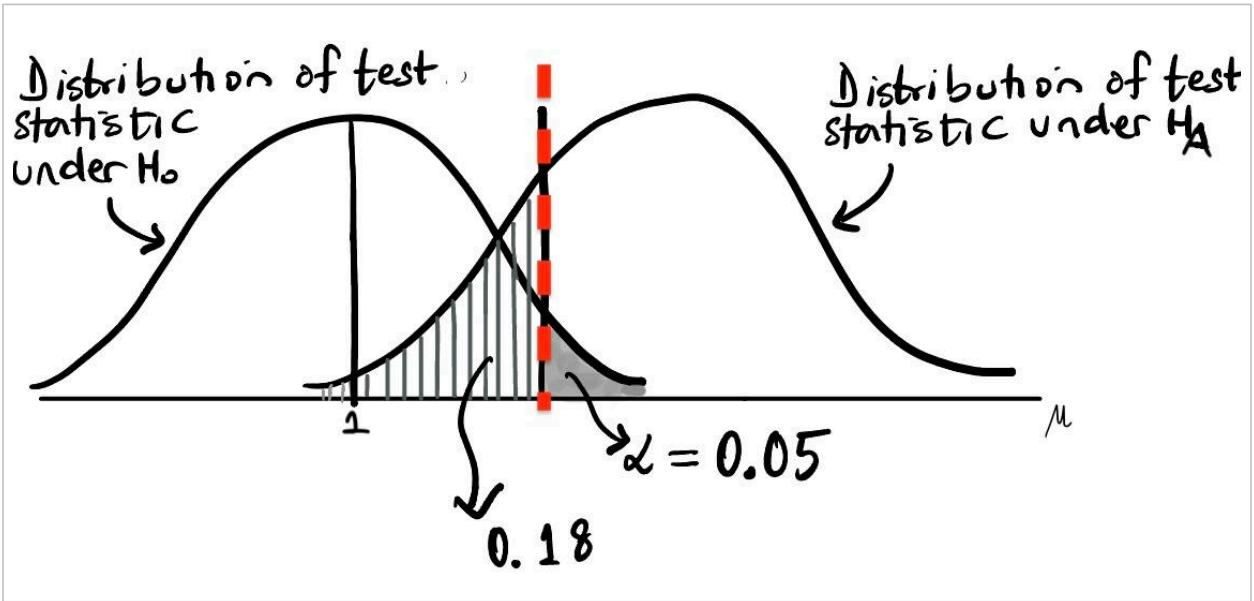
Warmup/Attendance: [Week 10: Friday 10/30/2026]

Full Name: SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

**1.** The following two questions are based on the diagram below. The diagram corresponds to a one-sided hypothesis test that uses the following null hypothesis: $H_0 : \mu = 1$, and a significance level of $\alpha = 0.05$. If $H_A$ represents a possible alternative hypothesis (an alternate distribution of the test statistic).




- What is an appropriate alternative hypothesis for this test based on the distribution under $H_A$ you see?

- How does a Type I error occur (under which distribution and on what side of the dashed line)?

- How does a Type II error occur (under which distribution and on what side of the dashed line)?

- Calculate the power of this test.